

P01

PROTOCOL 01

Privacy Layer for Solana — ZK-SNARKs, Stealth Addresses, Tor Relay,
Multi-Hop Routing, AI Agent

15

Programs

6

ZK Circuits

8

SDKs

56

AI Tools

Design & Architecture Document v0.9.1 — March 2026

Confidential — Volta Team

System Architecture

6-layer privacy stack from client to settlement. Every component is trustless, self-custody, and runs client-side.

CLIENT LAYER

MOBILE APP
React Native / Expo 54

EXTENSION
Chrome MV3 (beta)

WEB APP
Next.js 16

AI AGENT
56 Tools / On-Device LLM

PRIVACY LAYER

TOR RELAY
3-Tier IP Anonymization

AUTO-SHIELD
Stealth Recv + Auto Pool

PRIVACY ROUTER
Multi-Hop / 5 Levels

META-ADDRESSES
st:01 / st:02 P01-to-P01

SDK LAYER

specter-sdk
Stealth & Wallets

zk-sdk
Groth16 + STARK

zkspl-sdk
Confidential Bal.

arcium-sdk
MPC Compute

p01-js
Merchant SDK

privacy-toolkit
Merkle + Poseidon

auth-sdk
Auth Integration

rpc-config
RPC Fallback

PROTOCOL LAYER

STEALTH
ECDH + ML-KEM-768

SHIELDED
ZK Pool + Merkle

DENOMINATED
Fixed-Denom Pools

NOTE SPLIT
Cross-Pool Split

zkSPL
Confidential Bal.

PAYMENTS
Streams & Subs

VAULTS
Subscription Vaults

MPC
Arcium Cerberus

VERIFICATION LAYER

RELAYER
Trustless ZK

STARK
FRI + Goldilocks

QUANTUM
WOTS+ / HTL

BUNDLER
Atomic Batch

CRANK
Auto Payments

SOLANA BLOCKCHAIN

15 PROGRAMS
Anchor 0.32.1 / Rust

SPL Tokens
Token Standard

alt_bn128 + FRI
ZK Verification

Privacy Infrastructure

3-Tier Privacy Relay (Tor)

- \2713 Tier 1 (Client): fetchMiddleware strips User-Agent, Origin, Referer + 30-120ms jitter
- \2713 Tier 2 (Relay): Express server, HMAC rate limiter (zero IP storage), security headers
- \2713 Tier 3 (Tor): SOCKS5 proxy, circuit rotation every 10min via random auth
- \2713 Deployed on Railway — single container (Node 22 Alpine + Tor sidecar)
- \2713 13K+ requests tested, 100% Tor-routed, 0 errors

Auto-Shield Receive

- \2713 Receive screen generates one-time stealth addresses (HMAC from viewing secret key)
- \2713 Autonomous runner polls every 5min for incoming funds
- \2713 When balance detected (>0.05 SOL), auto-shields to best denomination pool
- \2713 Main wallet NEVER exposed to senders
- \2713 Settings toggle to enable/disable auto-shielding

Stealth Meta-Addresses (P01-to-P01)

- \2713 Format: st:01<base58(spending_ed25519 + viewing_x25519)>
- \2713 v2 hybrid: st:02 with ML-KEM-768 (quantum-resistant)
- \2713 Private Send auto-derives stealth destination from meta-address
- \2713 On-chain: stealth_sender → pool → stealth_receiver (no real wallet visible)

Multi-Hop Privacy Router

- \2713 5 privacy levels: Minimal (1 hop) → Paranoid (14+ hops, 20 splits)
- \2713 Autonomous runner with Android foreground service (survives app closure)
- \2713 Timing jitter: random delays between hops prevent correlation
- \2713 Note locking: notes in active routes cannot be withdrawn
- \2713 Consent popup: irreversibility warning before route starts
- \2713 Route progress: locked notes show hop X/Y + progress bar

Cryptographic Primitives

Stealth Addresses (ECDH)

- \2713 Ephemeral X25519 keypair per transaction
- \2713 ECDH shared secret + HKDF seed derivation
- \2713 View tags: 1-byte filter (99.6% quick rejection)
- \2713 v2: Hybrid X25519 + ML-KEM-768 (FIPS 203)
- \2713 On-chain registry (EIP-5564 for Solana)

Zero-Knowledge Proofs

- \2713 6 Groth16 circuits (BN254, up to 12,222 constraints)
- \2713 6 STARK AIRs (Winterfell/Goldilocks, quantum-resistant)
- \2713 note_split circuit (~10K constraints, max 20 outputs)
- \2713 Local-only proving via snarkjs (WebView on mobile)
- \2713 ~3s shield + unshield (filledSubtrees optimization)

Poseidon Hash

- \2713 BN254 scalar field (254-bit prime)
- \2713 $t=3$ (2 inputs), 63 rounds, $\alpha=5$
- \2713 Used for: note commitments, nullifiers, Merkle nodes
- \2713 poseidon-lite v0.3.0 (standardized everywhere)

Quantum-Safe Vault

- \2713 WOTS+ (Winternitz OTS): 32 hash chains, $w=16$
- \2713 Hash-Timelock: SHA-256 preimage + timelock
- \2713 Commit-Then-Reveal: 2-phase TX auth
- \2713 ML-KEM-768 for stealth v2

On-Chain Programs (15 Deployed on Devnet)

PROGRAM	PURPOSE	PROGRAM ID
zk_shielded	ZK pool, shield, unshield, split, denominated pools	6bVM5yve ... j27c
p01_zkspl	Confidential balances (zkSPL)	EqppogLB ... Ppah
specter	Stealth transfers + announcements	2tuztgD9 ... fbSp
p01_trustless	On-chain nullifier PDAs	FnTmMxsN ... i43Q
p01_relayer	Trustless ZK relay	2okhzLVr ... 5WpW
p01_quantum_vault	WOTS+, hash-timelock, commit-reveal	HazoS6VK ... Th7o
p01_registry	Stealth meta-address directory	QaQwpvBi ... hQB
p01_stark_verifier	Custom FRI verifier (6 circuits)	DGY37k3J ... QvDs
p01_arcium	9 MPC computation definitions	FH1JiQRU ... TLPT
p01_bundler	Atomic batch shield (up to 5/TX)	FzhzTRz8 ... F6GX
p01-fee-splitter	Hardcoded fee wallet	UdxXEvCA ... 5BM
subscription	Payment streams	(deployed)
stream	Real-time streaming	(deployed)
whitelist	Access control	(deployed)
p01_stark_verifier	Multi-circuit FRI (upgraded)	DGY37k3J ... QvDs

Security Model

Threat Model

- \2713 Blockchain observers cannot link senders and recipients
- \2713 Transaction amounts hidden in shielded transfers
- \2713 Spending patterns cannot be analyzed
- \2713 Balance tracking impossible for third parties

Guarantees

- \2713 Sound: invalid proofs cannot be generated
- \2713 Complete: valid spends always produce valid proofs
- \2713 Zero-knowledge: proofs reveal nothing beyond validity
- \2713 No double-spending: nullifiers unique per commitment
- \2713 MPC: 1-of-N honest node (Cerberus protocol)

Mobile Security Hardening

- \2713 Spending key never leaves device
- \2713 PIN: SHA-256 hashed, constant-time compare
- \2713 Progressive logout: 5→30s, 8→60s, 10→300s
- \2713 All secrets in hardware-backed SecureStore
- \2713 Clipboard auto-clear after 60s
- \2713 App switcher blur prevents screenshots
- \2713 android:allowBackup="false"
- \2713 Biometric auth with device fallback
- \2713 Lock screen enforced always

AI Agent — 56 Tools

On-device LLM (Gemma 3 via llama.rn). No data leaves the device. Tool-use loop: up to 3 rounds per message.

Wallet (6)

balance, address, send, history,
airdrop, stealth receive

Privacy (5)

shield, notes, private send,
route status, privacy level

Solana (6)

slot, epoch, TPS, tx lookup,
account info, rent calc

Token/DeFi (6)

token balance, swap quote,
execute, TVL, lending, portfolio

Analytics (3)

gas tracker, whale watch,
market dominance

Staking (3)

stake info, APY, validators

NFT (2)

list NFTs, floor price

Social (2)

protocol info, trending tokens

Conversion (3)

SOL/USD, SOL/lamports,
epoch/date

Device (3)

share, haptic, notification

Alerts (3)

price alerts set/list/clear

Utility (4)

calculate, validate, base58,
keypair

Technology Stack

Core

- Anchor 0.32.1, Solana CLI 2.2.14 (Agave)
- Rust 1.94.0, TypeScript 5.9.0
- React 19.1, Expo 54, RN 0.81, Next.js 16
- Node ≥22, pnpm 8.15, Turborepo

Cryptography

- snarkjs 0.7.5, circom2 v2.2.2
- Winterfell v0.10, Goldilocks field
- @noble/hashes, @noble/curves, @noble/post-quantum
- poseidon-lite v0.3.0, nacl (tweetnacl)

Infrastructure

- Railway (Tor relay), Vercel (web)
- Helius RPC (devnet), Docker + Tor sidecar
- Privy auth (embedded wallets)
- Jupiter aggregator (swaps)

Testing

- 370+ automated tests
- 0 TypeScript errors
- 56 screens, 80+ components audited
- vitest, ts-mocha, Rust tests